

Supplementary information

Benchmarking stimulus delivery and equilibration in cell-matched neuronal model replay

Supplementary scope

This supplement accompanies “Benchmarking stimulus delivery and equilibration in cell-matched neuronal model replay.” It provides prespecified recording-selection definitions, case-level results, metric applicability, numerical sensitivity results, model-construction mapping, and recording-unit verification. It does not introduce a pooled score, model ranking, preferred numerical condition, or replacement perisomatic result.

Supplementary Table S1: prespecified recording selections

Supplementary Table S1 lists all 24 recording selections. The first four rows describe the method-development specimen, with the original stimulus labels retained. The remaining 20 rows describe the four evaluation selections in each of five specimens. Whether any specific sweep was used during model fitting or evaluation could not be established; no selection is therefore described as held out or out of sample.

Supplementary Table S2: complete 72-case results

Supplementary Table S2 contains the 72 condition-specific cases, with raw spike counts, the three prespecified outcomes, applicability states, exact null reasons, usable coverage, and model identity. Rows 1–12 correspond to the method-development specimen, and rows 13–72 correspond to the five evaluation specimens. The ordering was prespecified and was not determined by outcome values.

The perisomatic rows reproduce the original implementation. Because no tested playback method satisfied all delivered-current benchmark waveforms and 20 of 24 cases met at least one material-change criterion in the equilibration sensitivity, these rows should not be interpreted as demonstrating physiologically faithful current delivery or an initialization-insensitive state.

Supplementary Table S3: complete 216-state metric applicability

Supplementary Table S3 contains three applicability entries for each of the 72 cases. Counts reconcile as follows:

Outcome	Calculated	Non-evaluable	Reason
Absolute spike-count error	72	0	unconditional
Conditional first-spike latency	20	52	one or both traces lacked a retained in-window spike
Long Square steady-state delta voltage	30	42	not applicable to 36 short protocols; insufficient model coverage in 6 cases

The six coverage-related non-evaluable outcomes are the three model conditions for specimen 471087830, Long Square sweeps 49 and 48. A missing numeric value together with its status and reason is the result; it is not equivalent to zero or to a failed comparison.

Outcome eligibility across 72 prespecified cases

Each row is one case; each cell is a prespecified outcome. Non-evaluable outcomes are not zero.

	Spike-count error	First-spike latency	Steady-state ΔV
01 314822529 sw 44 glif stored grid	C	SA	C
02 314822529 sw 44 glif recording grid	C	SA	C
03 314822529 sw 44 perisomatic recording grid playback	C	SA	C
04 314822529 sw 17 glif stored grid	C	SA	PS
05 314822529 sw 17 glif recording grid	C	SA	PS
06 314822529 sw 17 perisomatic recording grid playback	C	SA	PS
07 314822529 sw 75 glif stored grid	C	SA	PS
08 314822529 sw 75 glif recording grid	C	SA	PS
09 314822529 sw 75 perisomatic recording grid playback	C	C	PS
10 314822529 sw 66 glif stored grid	C	SA	C
11 314822529 sw 66 glif recording grid	C	SA	C
12 314822529 sw 66 perisomatic recording grid playback	C	C	C
13 324493977 sw 46 glif stored grid	C	SA	C
14 324493977 sw 46 glif recording grid	C	SA	C
15 324493977 sw 46 perisomatic recording grid playback	C	SA	C
16 324493977 sw 16 glif stored grid	C	SA	PS
17 324493977 sw 16 glif recording grid	C	SA	PS
18 324493977 sw 16 perisomatic recording grid playback	C	SA	PS
19 324493977 sw 69 glif stored grid	C	C	PS
20 324493977 sw 69 glif recording grid	C	C	PS
21 324493977 sw 69 perisomatic recording grid playback	C	C	PS
22 324493977 sw 37 glif stored grid	C	SA	C
23 324493977 sw 37 glif recording grid	C	SA	C
24 324493977 sw 37 perisomatic recording grid playback	C	SA	C
25 473943881 sw 41 glif stored grid	C	SA	C
26 473943881 sw 41 glif recording grid	C	SA	C
27 473943881 sw 41 perisomatic recording grid playback	C	SA	C
28 473943881 sw 16 glif stored grid	C	SA	PS
29 473943881 sw 16 glif recording grid	C	SA	PS
30 473943881 sw 16 perisomatic recording grid playback	C	SA	PS
31 473943881 sw 77 glif stored grid	C	C	PS
32 473943881 sw 77 glif recording grid	C	C	PS
33 473943881 sw 77 perisomatic recording grid playback	C	C	PS
34 473943881 sw 49 glif stored grid	C	SA	C
35 473943881 sw 49 glif recording grid	C	SA	C
36 473943881 sw 49 perisomatic recording grid playback	C	SA	C
37 471087830 sw 49 glif stored grid	C	C	SO
38 471087830 sw 49 glif recording grid	C	C	SO
39 471087830 sw 49 perisomatic recording grid playback	C	C	SO
40 471087830 sw 15 glif stored grid	C	SA	PS
41 471087830 sw 15 glif recording grid	C	SA	PS
42 471087830 sw 15 perisomatic recording grid playback	C	SA	PS
43 471087830 sw 90 glif stored grid	C	C	PS
44 471087830 sw 90 glif recording grid	C	C	PS
45 471087830 sw 90 perisomatic recording grid playback	C	C	PS
46 471087830 sw 48 glif stored grid	C	SA	SO
47 471087830 sw 48 glif recording grid	C	SA	SO
48 471087830 sw 48 perisomatic recording grid playback	C	SA	SO
49 475585413 sw 40 glif stored grid	C	SA	C
50 475585413 sw 40 glif recording grid	C	SA	C
51 475585413 sw 40 perisomatic recording grid playback	C	SA	C
52 475585413 sw 12 glif stored grid	C	SA	PS
53 475585413 sw 12 glif recording grid	C	SA	PS
54 475585413 sw 12 perisomatic recording grid playback	C	SA	PS
55 475585413 sw 70 glif stored grid	C	C	PS
56 475585413 sw 70 glif recording grid	C	C	PS
57 475585413 sw 70 perisomatic recording grid playback	C	C	PS
58 475585413 sw 32 glif stored grid	C	SA	C
59 475585413 sw 32 glif recording grid	C	SA	C
60 475585413 sw 32 perisomatic recording grid playback	C	SA	C
61 464188580 sw 39 glif stored grid	C	SA	C
62 464188580 sw 39 glif recording grid	C	SA	C
63 464188580 sw 39 perisomatic recording grid playback	C	SA	C
64 464188580 sw 19 glif stored grid	C	SA	PS
65 464188580 sw 19 glif recording grid	C	SA	PS
66 464188580 sw 19 perisomatic recording grid playback	C	SA	PS
67 464188580 sw 82 glif stored grid	C	C	PS
68 464188580 sw 82 glif recording grid	C	C	PS
69 464188580 sw 82 perisomatic recording grid playback	C	C	PS
70 464188580 sw 47 glif stored grid	C	SA	C
71 464188580 sw 47 glif recording grid	C	SA	C
72 464188580 sw 47 perisomatic recording grid playback	C	SA	C

Ccalculated

SAno in-window spike

PSnot applicable to protocol

SOinsufficient coverage

Supplementary Figure S1. Complete case-by-outcome eligibility matrix. Each row is one of 72 prespecified cases and each column is one outcome. Codes distinguish calculated values from non-evaluable outcomes caused by spike absence, protocol inapplicability, or insufficient coverage. The first 12 rows correspond to the method-development specimen.

Supplementary Table S4: evaluation-cell GLIF cases

Supplementary Table S4 contains all 40 GLIF cases from the five evaluation cells: 20 recording selections under two numerical conditions. The stored-grid condition precedes the recording-grid condition for presentation only. Input grid and numerical step change together, so the comparison does not isolate either factor and does not imply a preferred condition.

Supplementary Table S5: Short Square boundary sensitivity

Supplementary Table S5 provides the five complete event-boundary rows, including command indices, threshold and peak times, filtered derivatives, voltage at the right boundary, strict counts, post hoc counts, detector specification, and source-file identity. The prespecified half-open interval remains the primary estimand.

Supplementary Tables S6–S7: delivered-current validation benchmark

Supplementary Table S6 is the accessible 5×5 method-by-waveform matrix. Supplementary Table S7 provides transition, pulse, amplitude, charge, and edge-alignment results for all 25 combinations. No original candidate method satisfied all five waveforms, and none was applied to the 24 perisomatic cases.

This finding does not establish a replacement delivery method. A changed solver step, event-dispatch rule, or delivered-current estimand would define a new analysis and would require evaluation against the same benchmark set.

Supplementary Table S7A: post hoc constructive benchmark

Supplementary Table S7A reports the separately labeled, model-free constructive extension. Fixed steps of 0.005 and 0.0025 ms and an event-timed piecewise-constant reference met the unchanged criteria for all five waveforms. Fixed steps of 0.025 and 0.0125 ms met the complete criteria for three of five waveforms. These results do not establish a verified NEURON current-injection route, and no amended 24-case perisomatic analysis was run.

Supplementary Tables S8–S9: initialization/equilibration sensitivity

Supplementary Table S8 summarizes the cell-level findings. Supplementary Table S9 contains the 24 case-level reference and sensitivity comparisons, including exact reproduction, pre-stimulus stability, convergence time, spike counts, event shifts, steady-state shifts, and material-change status.

All 24 original pre-stimulus periods showed drift; every extended-equilibration run converged at 3.0 s; and 20 cases met at least one material-change criterion. Four Short Square cases remained zero/zero for stimulus-window spikes and did not meet another material-change criterion, although their original pre-stimulus periods were not stable. An independent calculation reproduced all 576 retained trajectory statistics and the two-window convergence classifications.

Supplementary Table S10: fit-section mapping

Supplementary Table S10 summarizes the six model constructions and all 158 section-targeting rules.

Specimen	Rules	Constructed sections	Constructed classes
314822529	28	79	soma 1; axon 2; basal dendritic 37; apical 39
324493977	26	65	soma 1; axon 2; basal dendritic 62
473943881	26	78	soma 1; axon 2; basal dendritic 75
471087830	26	36	soma 1; axon 2; basal dendritic 33
475585413	26	50	soma 1; axon 2; basal dendritic 47
464188580	26	47	soma 1; axon 2; basal dendritic 44

All 158 selectors mapped nonemptily and exclusively to the intended constructed-section class. The section counts describe the instantiated computational models and are not biological morphology comparisons.

Supplementary Table S11: recording units

Supplementary Table S11 reports file-specific response and stimulus units. Five NWB files use declared factor-1 conversion for response in volts and stimulus in amperes. Specimen 314822529, recording WKF 491202285, is a file-specific legacy exception: defective metadata conversion attributes are bypassed under the AllenSDK pre-1.1 identity rule, and stored numerical values are treated as SI. Display equations are $V \times 1,000$ for millivolts and $A \times 10^{12}$ for picoamperes.

This verification establishes units for the six named recordings. It does not evaluate recording quality, solver-time stimulus delivery, or biological validity.

Supplementary Table S12: interpretive scope

Analysis	Finding	Supported interpretation
Short Square boundary	Strict count 0/5 traces; post hoc +0.25 ms count 1/5 in every trace	Trace-specific primary result plus separately labeled boundary sensitivity
Delivered current	No original candidate satisfied all five benchmark waveforms	Numerical limitation of the tested playback routes; no claim of intended solver-time delivery
Initialization/equilibration	All original pre-stimulus periods showed drift; all extended-equilibration runs converged; 20/24 cases met a material-change criterion	Dependence on initialization/equilibration under the unchanged implementation
Fit-section mapping	All 158 selectors mapped to the intended constructed-section class	Exact construction-route identity only
Recording units	SI response and stimulus units verified for all six named NWB files	File-specific unit interpretation only
Combined interpretation	Results are descriptive and case-specific	No population inference, model ranking, held-out prediction, or broad biological validation

Software environment

Purpose	Recorded software identity
Scientific simulation and numerical sensitivities	micromamba 2.8.1; Python 3.11.15; NEURON 8.2.7; AllenSDK 2.16.2; NumPy 1.23.5; GCC 14.3.0; Ubuntu 26.04; GLIBC 2.43

The environments used during earlier recording acquisition and trace preparation were not fully versioned in the available records. No software version was inferred from a directory name.

Source metadata and animal reporting

The accompanying source metadata provide the six specimen IDs, retrieval time, exact Allen API v2 URLs, donor identifiers, sex, age, genotype, hemisphere, source structure, ephys-result IDs, and recording WKF IDs. The records confirm six distinct donors. No new animal procedure was performed for this secondary computational study.

Data and software availability

The public study materials contain metadata, derived tables, figures, and analysis software, but not Allen NWB files, model data, or compiled third-party binaries. Upstream objects must be retrieved from their original sources using the identifiers provided. Project-authored software is available under the MIT license.